DUNDALK INSTITUTE OF TECHNOLOGY

DK_ICSOP_7 - B.Sc. in Computing Systems and Operations

DK_ICCSO_8 – B.Sc. (Hons) in Computing Systems and Operations

Stage 3 - Semester 5

SAMPLE PAPER

COMP I8064 - Data Structures

Instructions

Answer QUESTION ONE (40 marks) and any other TWO questions (30 marks each).

You MUST attempt this question.

Question 1) Data Structure Fundamentals

(a) Data structures are a way of storing / organising information so that it can be used effectively and efficiently later.

(i) For each of the below data structures, give a real-world example where it would be a suitable choice. Provide a one sentence explanation for why this is the case.

- Binary Search Tree
- Queue
- List

(6 marks)

(ii) Give a real-world example of where each of the below data structures would be a poor choice. Provide a one sentence explanation for why this is the case.

- Stack
- Set
- HashMap

(6 marks)

(b) There are many data structures to choose from when programming. Identify two criteria that should be considered when choosing a data structure for use in an application. Explain why these criteria are of such importance.

(6 marks)

(c) Generics are a special form of data structure designed for flexibility and reuse.

(i) **Define** a generic data structure and **explain** how it differs from a non-generic data structure. Provide an example to illustrate your answer.

(5 marks)

(ii) Explain one benefit and one potential drawback to implementing a data structure as a generic version vs implementing using a strongly-typed approach.

(5 marks)

(d) Your team has been asked to write unit tests for a HashMap that manages collisions using collision chaining. The map holds String-User key-value pairs, with the key being the User's username (a unique and unchanging value). You have been assigned the following method:

a. public int put(String s, User u)

Outline three test cases that should be included as unit tests for the above method. For each test case, explain its necessity, what assertions should be included and why they are needed.

(12 marks)

Choose any TWO questions from questions 2, 3 and 4

Question 2) Binary Search Trees

(a) What is a “Binary Search Tree” data structure? List and explain the fundamental operations for a *Binary Search Tree*.

(10 marks)

(b) Write and explain the **pseudocode** for the *add* method in a Binary Search Tree. The Tree should contain String objects. Note: Your method should be done as pseudocode, not actual code. Pseudocode and explanation are worth equal marks.

(8 marks)

(c) Write and explain the **pseudocode** for the *removeMin* method in a Binary Search Tree. The Tree should contain String objects, and the removed value should be returned. Note: Your method should be done as pseudocode, not actual code. Pseudocode and explanation are worth equal marks.

(12 marks)

Question 3) Maps

(a) A Hash Map is a commonly used data structure that allows for fast data retrieval and storage. How is a Hash Map different from a standard array?

(4 marks)

(b) What is meant by the term *collision* in relation to a Hash Map? Explain the main causes of collisions in Hash Maps. What is the impact of *collisions* in a Hash Map without collision resolution?

(6 marks)

(c) The **load factor** is an important concept impacting Hash Map performance and efficiency.

(i) What is the load factor of a hash map? Explain how it is calculated and what it represents.

(6 marks)

(ii) How does a higher load factor impact the **performance** of a Hash Map? Your answer should include a discussion of the lookup & insertion actions.

(8 marks)

(iii) What is **rehashing**, and why is it necessary in Hash Maps? Explain the link between load factor and rehashing.

(6 marks)

Question 4) Analysis of Data Structures

(a) Your team has been assigned the task of building a ticketing system to store information on raised issues (stored as Ticket objects). You have been asked to research suitable data structures for the storage of Ticket information in memory (a database approach has been deemed unsuitable for this situation).

(i) Compare and contrast the suitability of the following structures as ticket storage for this system:

- Collision-resistant HashMap
- Tree-based Priority Queue
- Link-based Priority Queue

Your answer should include a discussion of time and space complexities. It should consider:

(i) Insertion.
(ii) Deletion.
(iii) Retrieval.

(27 marks)

(ii) Recommend, with reference to your answer to 4a) which structure should be employed if the system is to be deployed in a low memory environment where Tickets are frequently required to be retrieved in order of creation date.

(3 marks)